The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, AUTUMN SEMESTER 2021-2022

SYSTEMS AND ARCHITECTURE
(COMP 1030)

Time allowed ONE hour

An additional 30 minutes to cover IT issues and upload time

Answer TWO questions out of THREE

(only first two questions answers will marked)

Submit your answers containing all the work you wish to have marked as a **single PDF file, with each page in the correct orientation, to the appropriate drop box on the module's Moodle page.**

Use the standard naming convention for your document: [Student ID]_[Module Code]. Write your student ID number at the top of each page of your answers. Do not include your name.

Although you may use any notes or resources you wish to help you complete this open-book examination, the academic misconduct policy that applies to your coursework also applies here. You must be careful to avoid plagiarism, collusion or false authorship. This statement refers to, and does not replace, the University policy which stipulates severe penalties for academic misconduct.

Staff are not permitted to answer assessment or teaching queries during the period in which your open-book examination is live. If you spot what you think may be an error on the exam paper, note this in your submission but answer the question as written.

IMPORTANT: Lateness penalties will not be applied, and late submission will not be accepted.

1. Networks (25 Marks)

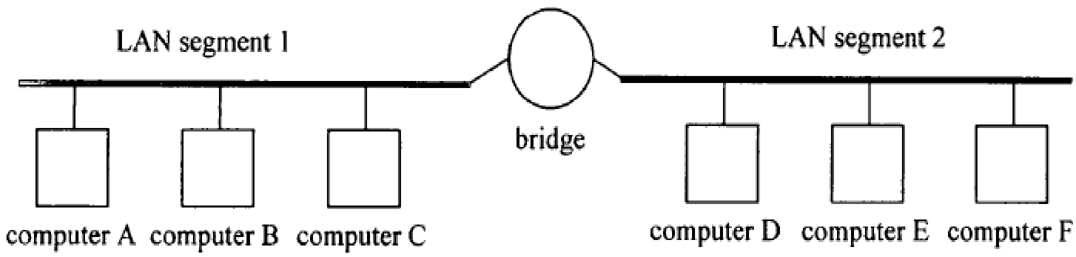
(a) Briefly explain the operations of the *CSMA/CA* protocol and explain why this protocol is needed for a wireless *LAN* access. [3 Marks]

(b) Describe each of the overhead factors in an Ethernet and a in a token ring Network and briefly explain how these factors affects the performance of a network.

[4 Marks]

(c) The Figure below shows a bridge that connects two LAN segments, each containing three computers. The table lists a sequence of events. Each event involves one of these computers transmitting a packet to another. This table also contains two columns to summarise the knowledge that the bridge has accumulated up to that point about which computers are on which segments. Assuming that the packets are transmitted in the order they appear in the table, write down a completed version of this table.

[6 Marks]



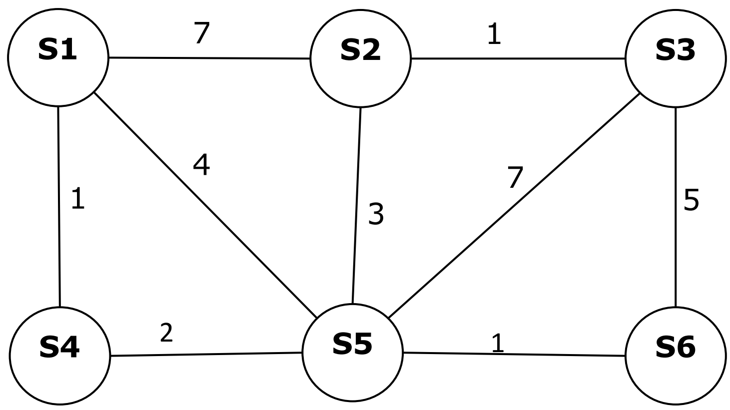
Event (transmission of a packet)	Knowledge of bridge about which computer is on segment 1	Knowledge of bridge about which computer is on segment 2
B sends packet to A		
D sends packet to F		
E broadcasts		
C sends packet to D		
A broadcasts		
F sends packet to B		

(d) You need to connect two buildings across a public road, and it is not feasible to use a direct cable connection. When would you use the following links?

(i) Leased line link

(ii) Satellite link [2 Marks]

(e) The graph in the Figure below summarises the connection between six packet switches that form the backbone of a wide area network, including the communication distances between them.



(i) Write down the route of the shortest path between switch **S3** and switch **S1**. Write down the total distance travelled along this path.

[1.5 Marks]

(ii) Write down the next hop routing table for Switch 2.

[2.5 Marks]

(f) One function of a routing protocol is to determine the "best" path between sending and receiving nodes. List and briefly explain any two criteria used to consider a "best" path?

[2 Marks]

(g) With reference to an Ethernet frame, explain how a host can support multiple network protocols via a single attachment to the physical transmission medium.

[2 marks]

(h) Describe two ways how the capacity of a LAN can be increased to accommodate heavier user data traffic.

[2 marks]

2. Binary arithmetic, Processor Registers and Addressing modes (25 Marks)

(a) The memory map below gives the contents of each of the locations of a simple 16-word memory. Each of these locations uses 4-bit addresses and contains a 4-bits binary value. Write the effect of executing each of the following ARM assembly language instructions.

0000	0010
0001	0011
0010	0010
0011	1010
0100	0000
0101	0010
0101	0001
0111	0011
1000	1010
1001	1111
1010	1010
1011	0011
1100	0001
1101	1000
1110	0000
1111	1010

Assume that r1 initially contains 0010 and r2 contains 1000

(i) `MOVE r0, #1110`

(ii) `LOAD r0, [r1]`

(iii) `LOAD r0, [r2]`

(iv) `LOAD r0, [r1, r2]`

(v) `LOAD r0, [r1, #5]`

[5 Marks]

(b) Some eight-bit microprocessors and classic CISC computers have variable-length instructions.

(i) Write the advantages of computer architectures with instructions of different lengths?

(ii) What are the disadvantages?

[4 Marks]

(c) Represent the number $(0.625)_{10}$ in the following single precision and double precision formats. The exponent is in excess 127 and excess 1023 respectively.

Single Precision	**Double Precision**
Sign ← 1 bit	Sign ← 1 bit
Exponent ← 8 bits	Exponent ← 11 bits
Mantissa ← 23 bits	Mantissa ← 52 bits

[4 Marks]

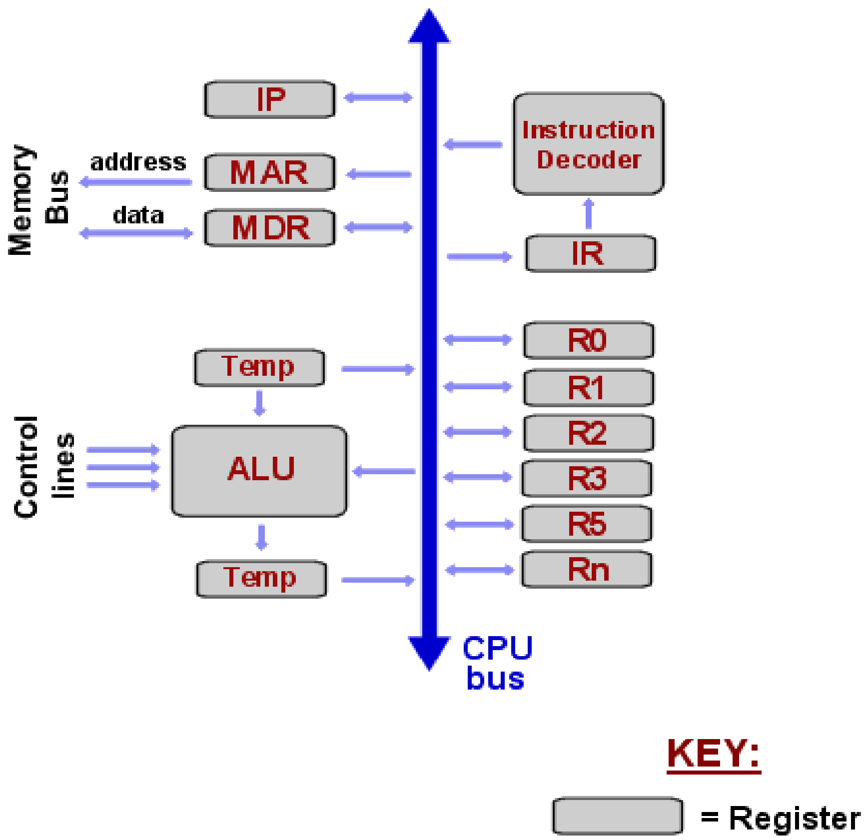
(d) Multiply the decimal number 31 by 21 in binary using a binary multiplication algorithm. Initially, the multiplicand is in *Y(5)* register and the multiplier in *B(5)* register. The sequence counter *SC* is set to a number equal to the number of bits of the multiplier. The accumulator *A(5)* is used to do the arithmetic operation and an extra carry bit of an accumulator will be transferred to a register *E(1)*. The value given in bracket is the size of each register. Write the steps needed for each operation and show the contents of registers *E*, *B, Y*, *A*, and *SC* during each stage of the multiplication process.

[6 Marks]

(e) Suppose that the programming instructions are executed by the following components as shown in the following diagram. Write the complete control steps needed for execution of the following instruction.

ADD (R3), R1

[6 Marks]



3. Interrupts, I/O, Operating System and Instruction Pipelining (25 Marks)

(a) Consider the interrupt that occurs at the completion of a data transfer from the disk to CPU.

(i) Why is the interrupt used in this case? (1)

(ii) What would be necessary if there were no interrupt capability on this computer? (2)

(iii) Describe the steps that take place after the interrupt occurs. (2)

[5 Marks]

(b) (i) What is polling used for? (2)

(ii) What are the disadvantages of polling? (2)

(iii) Upon an interrupt what has to happen implicitly in hardware before control is transferred to the interrupt handler? (3)

[7 Marks]

(c) Identify the WAR, WAW and RAW data dependencies in the following piece of code.

```
L1: R1 = 50
L2: R1 = R3 + R4
L3: R2 = R4 - 10
L4: R4 = R1 + R2
L5: R2 = R1 + 45
```

[5 Marks]

(d) A certain disk interface accepts requests to read a 32 KB block of data. It has a 32 KB buffer on board for its I/O interface, in which it stores the data as it comes off the drive. The interface is interrupt-driven and has DMA capability. Describe the likely sequence of events from the time the processor requests a block of data until the data have been transferred to the main memory.

[4 Marks]

(e) List and briefly explain any four major functions of an operating systems with examples.

[4 Marks]